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Task Description:	Humans exposed to prolonged periods of microgravity undergo deconditioning of the cardiovascular system and exhibit increased incidence of orthostatic intolerance. In addition, the effects of cardiovascular deconditioning seem to be greater in females as compared to males that undergo periods of spaceflight. Hindlimb unloading (HU) in rodents is a well described animal model used to simulate exposure to microgravity in humans. HU rats exhibit signs of deconditioning including effects on the cardiovascular system that are very similar to those experienced by humans following spaceflight. The arterial baroreflex is a major component in the beat to beat maintenance of blood pressure and is critical for the normal compensation to an orthostatic stress. Following HU, baroreflex mediated increases in sympathetic nerve activity in response to hypotensive stimuli are attenuated, and this seems blunted to a greater extent in female rats following HU. This change in autonomic function is due in part to an increase in the effects of the inhibitory neurotransmitter gamma-aminobutyric acid (GABA) in a specific nucleus of the hindbrain, rostral ventrolateral medulla (RVLM), which is important in arterial baroreflex function and control of sympathetic activity. In addition, GABA(A) receptors are complex multimeric proteins that have a range of function and susceptibility to positive modulators is determined by their subunit composition. Taken together, the increased GABA neurotransmission in the RVLM following HU could be mediated by GABA receptors that are potentiated either due to changes in expression of individual subunits, presence of positive modulatory substances, enhanced susceptibility to endogenous modulators, increased number of GABA(A) receptors, greater GABA release, or a combination of these possibilities. The current proposal will test the general hypothesis that simulated microgravity alters GABA(A) neurotransmission in the RVLM, and to a greater extent in females, which accounts for the changes in autonomic function in both males and females. The project has 4 specific objectives: 1. To examine the effects of HU on GABA(A) and GABA(B) receptor mediated inhibition in the RVLM in male and female rats. 2. To investigate alterations in GABA(A) receptor sensitivity to neurosteroids following HU in male and female rats. 3. To determine if HU in male and female rats alters nitric oxide neurotransmission and/or positive modulation of GABA(A) mediated inhibition in the RVLM. 4. To determine if known endogenous positive modulators of GABA(A) neurotransmission (nitric oxide and neurosteroids) contribute to autonomic changes that occur following HU in male and female rats. Cardiovascular deconditioning results in reduced exercise capacity and orthostatic intolerance and these effects are more pronounced in women compared to men. Although a number of mechanisms likely are involved in the predisposition for orthostatic intolerance, dysfunction of cardiovascular sympathetic regulatory mechanisms appears to have an important contribution. Sex differences exist in arterial baroreflex control of the sympathetic nervous system and this reflex is altered following HU in male rats. Blunted baroreflex sympathoexcitation in male HU rats appears to be due to a central nervous system mechanism. Little is known concerning central mechanisms that may contribute to sex differences in cardiovascular deconditioning. Using the hindlimb unloaded rat model for cardiovascular deconditioning, the proposed studies will examine sex differences in GABA neurotransmission in the RVLM, the nucleus that is the major determinant of efferent sympathetic outflow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Control of the autonomic nervous system is altered during by exposure to microgravity or prolonged bedrest. This could contribute to the increased incidence of orthostatic hypotension and reduced ability to maintain blood pressure during hemorrhagic events following cardiovascular denconditioning. In addition, evidence suggests that females are more susceptible than males to these problems following spaceflight or prolonged bedrest. Defining the central nervous system mechanisms that account for the changes in autonomic nervous system function following cardiovascular deconditioning will increase our understanding of the adaptations that occur in males and females following spaceflight. These studies utilize a ground base model of cardiovascular deconditioning. Therefore, the knowledge gained from these studies would apply not only to exposure to midcrogravity but also to patients exposed to prolonged bedrest and to normal neural control of the circulation.
	During this period, progress has been made on several difference research projects investigating the cardiovascular changes following HU. These include studies in our laboratory and collaborative projects with other investigators. Three manuscripts were accepted for publication during this time period. We have submitted an additional manuscript that is in review. In addition, we have initiated studies investigating the effects of HU both on the neuronal and inducible nitric oxide synthase immunohistochemistry in the RVLM and on GABAA neurotransmission in the RVLM. Furthermore, we have contributed to several joint projects with other researches summarized below. Successful adaptations to the traditional hindlimb suspension model in the rat. The purpose of this study was to evaluate an alternative method to the Morey-Holton technique for hind limb unloading (HU) that would permit tail examination during long-term suspension, surgery during unloading and a seamless recovery phase following HU. The proposed method also provides a solution to the problem of HU rats chewing the traction tape and coming out of the suspension apparatus. Data are presented from two separate laboratories currently using the new technique. Young (9 weeks) male and female (n=12) and mature (6 and 16 months) female rats (n=12) were HU for 2 or 4 weeks with or without 2 weeks of recovery. Two stainless steel wires were passed through separate intervertebral disc spaces at the level of the ~5th and 7th sacral vertebrae and fashioned into a ring from which animals were suspended. This position was chosen as it is not too proximal to interfere with defecation and not too distal where vertebrae are too small to hold body weight. All HU groups had ambulatory controls, some of which were pair-fed. Adrenal weights, daily food consumption, body weight lost during suspension and atrophy of the soleus muscle were used to determine success of hind limb suspension. Body mass was not significantly different at baseline and no weight loss occurred after unloading in any of the groups. Soleus muscle mass was significantly reduced with unloading ($p<0.001$) in all groups. Food consumption tended to be higher in the HU groups compared to controls and thus, there was no food restriction of controls. Adrenal gland weights did not differ between controls and any of the HU rats. The lack of significant change in body mass, food consumption, adrenal gland weight, coupled with significant soleus muscle atrophy, suggest this method is as successful as the traditional Morey-Holton technique. The method described confers several advantages such as ease of reloading and elimination of rats chewing the traction tape and becoming ambulatory. The disadvantage to the technique is that rats must be anesthetized for 5-7 minutes to implant tail rings. Increased nitric oxide synthase in the hypothalamus of hindlimb unloaded rats. Upon return from spaceflight or resumption of normal posture after bedrest, individuals often exhibit cardiovascular deconditioning. Cardiovascular deconditioning is characterized by reduced exercise capacity, resting tachycardia, and in some cases, orthostatic

intolerance. Although the mechanisms responsible for cardiovascular deconditioning have yet to be fully elucidated, alterations within the central nervous system have been postulated to be involved. The paraventricular nucleus (PVN) and supraoptic nucleus (SON) of the hypothalamus are important brain regions in control of sympathetic outflow and body fluid homeostasis. Nitric oxide (NO) regulates the activity of PVN and SON neurons, and alterations in NO transmission within these brain regions may contribute to symptoms of cardiovascular deconditioning. The purpose of the present study was to examine nitric oxide synthase (NOS) in the PVN and SON of control and hindlimb unloaded (HU) rats, an animal model of cardiovascular deconditioning. The number of neuronal NOS (nNOS) positive neurons, as assessed by NADPH diaphorase staining, was significantly elevated in the PVN but not SON of HU rats. HU rats had a significantly greater number of NADPH-diaphorase positive neurons suggesting that cardiovascular deconditioning produces an increase in NOS in the PVN. Western blot analysis revealed that nNOS but not endothelial NOS (eNOS) protein expression was higher in the PVN of HU rats. In the SON, there was strong trend for an increase in nNOS ($p=0.052$) and a significant increase in eNOS expression in HU rats. Our results indicate that NOS expression is increased in the PVN and SON by hindlimb unloading and may contribute to alterations in autonomic function and body fluid regulation following periods that induce cardiovascular deconditioning.

Effects of sex and HU on arterial baroreflex function. Orthostatic intolerance is a concern in astronauts following spaceflight and the incidence of presyncope post-spaceflight is much greater in female compared to male astronauts. Normal compensation to changes in posture requires appropriate neurohumoral responses. There is evidence that attenuated arterial baroreflex sympathoexcitation and inadequate vasoconstriction contribute to reduced orthostatic tolerance in astronauts post-flight and in ground-based models of microgravity in humans and animals. Differences in baroreflex mediated responses to posture in females compared to males may account for the greater incidence of orthostatic intolerance in female astronauts. This study evaluated the effects of HU on arterial baroreflex function in male (M) and female (F) rats. Arterial baroreflex responses in heart rate and renal sympathetic nerve activity (RSNA) to increases and decreases in mean arterial pressure (MAP) produced by intravenous infusion of phenylephrine and nitroprusside, respectively were recorded in conscious animals. Baroreflex control of heart rate was very similar with only the maximum heart rate in response to decreases in MAP being greater in the HU as compared to the control animals of both sexes. This is consistent with the resting tachycardia seen following HU. Sex (females<males) and HU attenuated baroreflex mediated sympathoexcitation. This additive effect in the female HU results in the greatest reduction in sympathetic reserve and could contribute to the greater incidence of orthostatic intolerance in females following cardiovascular deconditioning.

Task Progress:

Hindlimb unloading alters nitric oxide and autonomic control of resting arterial pressure in conscious rats. After periods of microgravity or bed rest, individuals often exhibit reduced VO2 max, hypovolemia, cardiac and vascular effects, and autonomic dysfunction. Recently, alterations in expression of vascular and central nervous system NO synthase (NOS) have been observed in hindlimb-unloaded (HU) rats, a model used to simulate physiological effects of microgravity or bed rest. We examined the effects of 14 days of hindlimb unloading on hemodynamic responses to systemic NOS inhibition in conscious control and HU rats. Because differences in NO and autonomic regulation might occur after hindlimb unloading, we also evaluated potential differences in resting autonomic tone and effects of NOS inhibition after autonomic blockade. Administration of nitro-L-arginine methyl ester (L-NAME; 20 mg/kg iv) increased mean arterial pressure (MAP) to similar levels in control and HU rats. However, the change in MAP in response to L-NAME was less in HU rats that had an elevated baseline MAP. In separate experiments, atropine (1 mg/kg iv) increased heart rate (HR) in control but not HU rats. Subsequent administration of the ganglionic blocker hexamethonium (30 mg/kg iv) decreased MAP and HR to a greater extent in HU rats. Administration of L-NAME after autonomic blockade increased MAP in both groups to a greater extent compared with intact conditions. However, the pressor response to L-NAME was still reduced in HU rats. These data suggest that hindlimb unloading in rats reduces peripheral NO as well as cardiac parasympathetic tone. Along with elevations in sympathetic tone, these effects likely contribute to alterations in vascular control and changes in autonomic reflex function following spaceflight or bed rest.

Interaction between GABA and excitatory amino acids in control of sympathetic nerve activity by the rostral ventrolateral medulla in hindlimb unloaded rats. Previously we have reported that pressor and sympathoexcitatory responses to blockade of ?-amino butyric acid (GABAA) receptors in the rostral ventrolateral medulla (RVLM) are enhanced in hindlimb unloaded (HU) rats, a model of cardiovascular deconditioning. Excitatory amino acids (EAAs) may play a role in the sympathoexcitation produced by removal of GABAergic tone in the RVLM. Therefore, we hypothesized that EAAs contribute to enhanced responses to GABAA blockade observed in HU rats. To test this hypothesis, rats were maintained under HU or control conditions for 14 days. Bilateral RVLM microinjections were then performed under Inactin anesthesia while recording mean arterial pressure (MAP) and lumbar sympathetic nerve activity (LSNA). The EAA antagonist kynurenate (40 mM, 90 nl) had little or no effect alone in HU or control rats but attenuated increases in MAP and LSNA produced by bicuculline (5 mM, 90nl) in both groups. However, the response to bicuculline in the presence of kynurenate was enhanced in HU versus control rats. These data suggest that although EAAs play a role in the pressor and sympathoexcitatory responses to GABAA receptor blockade in the RVLM, they do not appear to account for differences observed in control and HU rats. We speculate that differences in GABAA tone and/or non-EAA mediated transmission are responsible for enhanced responses to bicuculline in rats following a period of cardiovascular deconditioning.

Expression of GABAA receptor subunits and NOS in the RVLM. The purpose of these studies is to examine the effects of hindlimb unloading (HU) on mRNA and protein expression of different neurotransmitter signaling systems in the RVLM in male and female rats. Previous studies have shown that GABAA mediated inhibition in the RVLM is increased following HU in male rats and this could be due to increased positive modulation by neurosteroids. Neurosterods are metabolites of progesterone and are potent endogenous positive modulators of GABAA receptor function. The neurosteroids directly bind to the receptor and facilitate the actions of GABA. The ability of neurosteroids to positively modulate GABAA receptors is dependent on the subunit composition of the GABAA receptors. Following HU, enhancement of GABAA receptor function due to neurosteroids may be greater in females than males. Potentiation of GABAA receptors by neurosteroids is consistent with enhanced GABAA mediated inhibition in the RVLM following HU and may be a mechanism that accounts for the greater reduction in baroreflex mediated sympathoexcitation. These studies will evaluate changes in expression in GABAA receptor subunits important in determining neurosteroid sensitivity. To date, we have developed the standards and optimized the real time reverse transcriptase-polymerase chain reaction (RT-PCR) necessary to quantify mRNA expression of glyceraldehydes 3-phosphate dehydrogenase (GAPDH) and GABAA alpha1, alpha2, gamma1, gamma2, gamma3, and delta receptor subunits. We are currently working on cloning and producing the standards for GABAA alpha3, alpha4, alpha5, and alpha6 receptor subunits. With the development of these techniques, we can now investigate differences in the expression of specific GABAA receptor subunits in the RVLM between male and female rats and those produced by a period of cardiovascular deconditioning.

	In addition to GABAA receptor subunits, we have developed the techniques to evaluate both mRNA and protein expression of different nitric oxide synthase (NOS) isoforms using both real time RT-PCR (as above) and immunoblots. In addition, we have developed immunocytochemistry methods to detect NOS isoforms in order to localize the specific areas that are affected by HU. To further determine which neuronal populations are altered the NOS immunocytochemistry will be combined with retrograde labeling and immunocytochemistry for other phenotypic markers. For example, the retrograde labeling in the spinal cord will label neurons in the RVLM that project to the spinal cord and control sympathetic nerve activity. In addition, staining for catecholamine enzymes (tyrosine hydroxylase and/or phenyl ethanolamine-n-methyltransferase) will be utilized to differentiate the phenotype of neurons in the RVLM. Combining (double and triple labeling) the NOS isoform staining and retrograde/catecholamine label will allow for determination of the changes in NOS expression following HU to be localized to specific neuronal populations in the RVLM. This information would be critical in designing physiologic models and be a foundation to understand the changes that occurs following HU and lead to rationale development of countermeasures.
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